

# FORCE AND LAWS OF MOTION

Complete Concept Notes & Formula Sheet — Class 9 NCERT Physics

Comprehensive Guide

CBSE / NCERT

Physics Notes

## 1. INTRODUCTION TO FORCE

A **force** is a push or pull upon an object resulting from the object's interaction with another object. Whenever there is an interaction between two objects, there is a force upon each of the objects.

### EFFECTS OF FORCE

Forces can produce several effects on a body:

- Move a stationary body or stop a moving body.
- Change the speed of a moving body (accelerate/decelerate).
- Change the direction of motion of a body.
- Change the shape and size of an object (e.g., stretching a spring, squeezing a rubber ball).

## 2. BALANCED AND UNBALANCED FORCES

| Parameter                               | Balanced Forces                                      | Unbalanced Forces                                         |
|-----------------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| <b>Net Force (<math>F_{net}</math>)</b> | Equal to zero ( $F_{net} = 0$ ).                     | Not equal to zero ( $F_{net} \neq 0$ ).                   |
| <b>State of Motion</b>                  | Does not change the state of rest or uniform motion. | Changes the speed or direction of motion.                 |
| <b>Acceleration</b>                     | Produces zero acceleration ( $a = 0$ ).              | Produces a net acceleration ( $a \neq 0$ ).               |
| <b>Shape</b>                            | Can change the shape/size of the object.             | Maintains or changes shape while causing motion.          |
| <b>Example</b>                          | Tug-of-war when both teams pull with equal force.    | Tug-of-war when one team pulls harder and moves the rope. |

## 3. INERTIA AND FIRST LAW OF MOTION

### Newton's First Law of Motion (Law of Inertia)

An object remains in a state of rest or of uniform motion in a straight line unless acted upon by an external unbalanced force.

### Inertia & Mass

**Inertia** is the inherent property of an object to resist any change in its state of rest or uniform motion. The **mass** of a body is a measure of its inertia. Heavier objects have higher inertia.

## Types of Inertia

- **Inertia of Rest:** Inability of a body to change its state of rest by itself.

*Example:* Leaves fall when a tree branch is shaken vigorously; passengers jerk backward when a bus suddenly starts.

- **Inertia of Motion:** Inability of a body to change its state of uniform motion by itself.

*Example:* Passengers jerk forward when a bus suddenly stops; an athlete runs a distance before jumping.

- **Inertia of Direction:** Inability of a body to change its direction of motion by itself.

*Example:* Mud flies off tangentially from a rotating vehicle tire; passengers tilt sideways when a car takes a sharp turn.

## 4. MOMENTUM

**Momentum ( $p$ )** is defined as the product of the mass ( $m$ ) of an object and its velocity ( $v$ ). It is a vector quantity having both magnitude and direction (same direction as velocity).

$$p = m \times v$$

- **SI Unit:** Kilogram-meter per second ( $\text{kg}\cdot\text{m/s}$  or  $\text{kg m s}^{-1}$ ).
- If a body is at rest ( $v = 0$ ), its momentum is **zero**.

## 5. NEWTON'S SECOND LAW OF MOTION

### Newton's Second Law of Motion

The rate of change of momentum of an object is directly proportional to the applied unbalanced force in the direction of the force.

### Mathematical Derivation

Let an object of mass  $m$  move along a straight line with an initial velocity  $u$ . When a constant force  $F$  acts on it for time  $t$ , its velocity becomes  $v$ .

- Initial momentum ( $p_1$ ) =  $m \times u$
- Final momentum ( $p_2$ ) =  $m \times v$
- Change in momentum ( $\Delta p$ ) =  $p_2 - p_1 = m(v - u)$
- Rate of change of momentum =  $m(v - u) / t$

According to the Second Law:

$$F \propto m(v - u) / t \Rightarrow \text{Since } a = (v - u) / t, \quad F \propto m \times a$$

$$F = k \times m \times a \quad (\text{Taking constant } k = 1) \Rightarrow F = m \times a$$

## SI UNIT OF FORCE

The SI unit of force is **Newton (N)** or  $\text{kg}\cdot\text{m/s}^2$ .

**1 Newton Definition:** The force required to produce an acceleration of  $1 \text{ m/s}^2$  on an object of mass  $1 \text{ kg}$ .

$$1 \text{ N} = 1 \text{ kg} \times 1 \text{ m/s}^2$$

## Applications of Second Law

- **Catching a Cricket Ball:** A fielder pulls his hands backward while catching a fast ball. Increasing the time ( $t$ ) decreases the rate of change of momentum, reducing the impact force ( $F$ ) on hands.
- **High Jump Athletes:** Athletes land on a cushioned bed or sand pit to increase time of impact, thereby reducing the force of landing and preventing injury.
- **Seat Belts in Cars:** Seat belts stretch slightly to increase the deceleration time during collision, reducing the force exerted on passengers.

## 6. NEWTON'S THIRD LAW OF MOTION

### Newton's Third Law of Motion

To every action, there is an equal and opposite reaction, and they act on two different bodies.

$$F_{A \text{ on } B} = -F_{B \text{ on } A}$$

### KEY CHARACTERISTICS OF ACTION-REACTION PAIRS

- Action and reaction forces act **simultaneously**.
- They always act on **two different objects**, so they never cancel each other out.
- They are equal in magnitude but opposite in direction.

## Examples & Applications

- **Walking on the Ground:** We push the ground backward (action), and ground pushes us forward (reaction).
- **Recoil of a Gun:** When a bullet is fired from a gun, it exerts a forward force on the bullet (action), and the bullet exerts an equal backward force on the gun (recoil reaction).
- **Rocket Propulsion:** Expanding gases escape downward at high velocity (action), pushing the rocket upward (reaction).
- **Rowing a Boat:** Pushing the water backward with oars propels the boat forward.

## 7. LAW OF CONSERVATION OF LINEAR MOMENTUM

### Law of Conservation of Momentum

In an isolated system (where no external unbalanced force acts), the total momentum before collision is equal to the total momentum after collision.

## Mathematical Proof

Consider two balls A and B of masses  $m_A$  and  $m_B$  moving in the same direction along a straight line with initial velocities  $u_A$  and  $u_B$  (where  $u_A > u_B$ ).

- During collision lasting for time  $t$ , ball A exerts force  $F_{AB}$  on B, and ball B exerts force  $F_{BA}$  on A.
- Final velocities after collision are  $v_A$  and  $v_B$ .

From Newton's Second Law:

Force on ball B by A:  $F_{AB} = m_B(v_B - u_B) / t$

Force on ball A by B:  $F_{BA} = m_A(v_A - u_A) / t$

According to Newton's Third Law ( $F_{AB} = -F_{BA}$ ):

$$m_B(v_B - u_B) / t = -m_A(v_A - u_A) / t$$

$$m_B v_B - m_B u_B = -m_A v_A + m_A u_A$$

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

**Total Initial Momentum = Total Final Momentum**

## Recoil Velocity of a Gun

Let mass of gun =  $M$ , mass of bullet =  $m$ .

Initial velocity of both before firing = 0.

Velocity of bullet after firing =  $v$ , recoil velocity of gun =  $V$ .

$$0 = M \cdot V + m \cdot v \Rightarrow V = -(m \cdot v) / M$$

## 8. SUMMARY FORMULA QUICK-REFERENCE

| Concept                  | Formula                                 | Variables & Units                                  |
|--------------------------|-----------------------------------------|----------------------------------------------------|
| Momentum                 | $p = m \cdot v$                         | $m$ = mass (kg), $v$ = velocity (m/s)              |
| Newton's 2nd Law         | $F = m \cdot a$                         | $F$ = force (N), $a$ = acceleration ( $m/s^2$ )    |
| Acceleration             | $a = (v - u) / t$                       | $u$ = initial vel, $v$ = final vel, $t$ = time (s) |
| Conservation of Momentum | $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$ | Subscripts 1, 2 denote bodies 1 and 2              |
| Recoil Velocity          | $V_{recoil} = -(m \cdot v) / M$         | $M$ = mass of gun, $m$ = mass of bullet            |